HEART AND METAL — PRESS & CREATOR KIT

Project Overview & Technical Summary

1. About the Project

Specification	Detail
Game Title	Heart and Metal
Genre	Interactive 3D Puzzle / Action Title
Engine	Unity Engine (Universal Render Pipeline)
Target Platform	PC
Project Role	Technical Sound Designer & Audio Lead (Jacob Barrett)

2. Creator's Statement: The "Why" and "How"

As the Technical Sound Designer on Heart and Metal, my goal was to embed environmental sound effects and craft a deeply responsive space. In game development, audio is frequently treated as an afterthought, a layer added on top of finished gameplay. For Heart and Metal, the objective was for the soundscape to actively reinforce narrative tension, mechanical feedback, and spatial awareness throughout the player's journey.

To achieve robust ambient environmental management across a multi-player co-op architecture where spatial rooms and portals presented scaling hurdles, I integrated Wwise's dependable AkAmbience system. This allowed dynamic, zone-based ambient beds and transitions to wrap smoothly around players as they navigate industrial level geometry.

Furthermore, to eliminate repetitive audio fatigue, I built custom C# raycast-driven footstep and surface detection systems that dynamically interrogate underlying terrain textures, matching audio material instantly to concrete or metal grates.

Asset sourcing focused strictly on curated, high-grade royalty-free sound libraries. Every environmental emitter, mechanical pulse, and impact effect was selectively gathered from premier royalty-free sound collections, then re-edited, layered, and routed through specialized sub-busses. This disciplined asset curation ensured that

dual-listener co-op perspectives remained acoustically distinct, cohesive, and completely free of muddy masking.

3. Technical Specifications & Audio Pipeline

- **Audio Middleware:** Wwise (Audiokinetic) integrated with Unity Engine (URP)
- **Spatial & Ambient Architecture:** Wwise AkAmbience system for reliable zone-based environmental bed management and smooth multi-player audio transitions
- **Scripting & Integration:** Custom C# raycasting for surface-dependent footstep and interaction detection across diverse terrain materials
- **Mixing & Routing:** Multi-bussing architecture featuring dedicated sub-routes for mechanical hazards and environmental ambiences
- **Asset Sourcing & Implementation:** Meticulous curation and integration of specialized royalty-free sound effects, ambient beds, and multi-layered impact designs for metallic structures, water flows, and interactive puzzle elements

4. Stills











